DIFFERENTIATED CO-DESIGN COMPILER

VQETape

Compile the forward contraction, reverse program, and variational ansatz as one optimization problem.

8.2% faster amortized objective VQ spatial vs TC-NG	28.3% less host peak RSS 473.9 vs 661.3 MiB	3.38 ms VQ warm frontier 2.66 ms TC-NG reference

DEMONSTRATED RESULT

On one matched RTX 3090, VQETape spatial transfer records 16.7781 s for compile + first + 100 warm calls versus 18.2720 s for TensorCircuit-NG, an 8.2% improvement. Host peak RSS moves from 661.3 to 473.9 MiB, a 28.3% reduction. The measured warm reference and validated Fig. 2 construction define two precise scale-forward targets.

Team Ranger - Junkai Wang
Challenge #33 - PR #263
Evidence snapshot: 748c5e974573
30 July 2026

1. From contraction order to differentiated-program co-design

TensorCircuit-NG supplies a tensor-native computational graph, automatic differentiation, contraction-path optimization, slicing, and distributed execution. VQETape adds a complementary compiler layer: the reverse program, saved residuals, checkpoint schedule, exact symmetry sector, optimizer, and ansatz growth enter the same measured search space as the forward contraction.

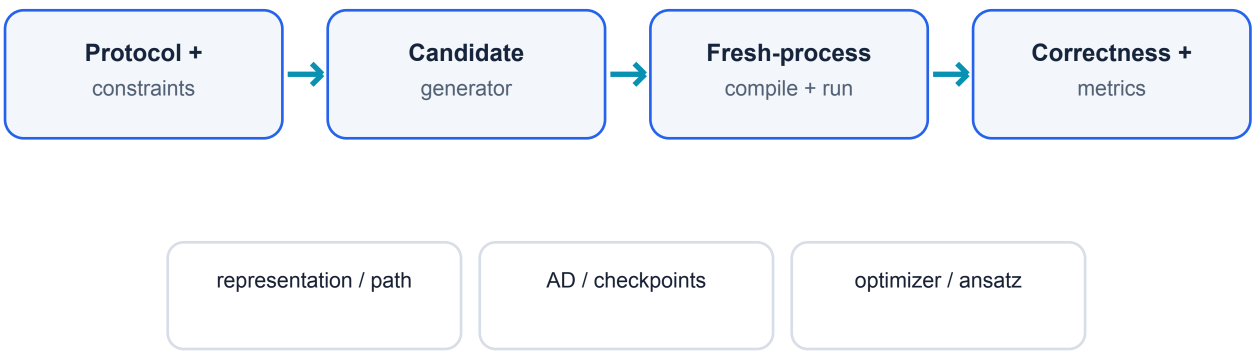
Matched physical protocol

Open-boundary transverse-field Ising model: $H = -J \,\mathrm{sum}\, Z_i Z_{(i+1)} - g \,\mathrm{sum}\, X_i$, with $J=g=1$. The matched run uses $n=10$, depth $L=4$, the plus initial state, RZZ then RX per layer, seed 33, complex64, and five synchronized warm repetitions on one NVIDIA RTX 3090.

Selection objective

$T_{objective} = T_{compile} + T_{first} + 100 \times \mathrm{median}(T_{warm})$. Compilation includes measured path search. Energy and the complete gradient must satisfy 1e-5 tolerances before selection.

Compiler feedback loop



Auto-evaluation feeds measured cost and validity back into selection

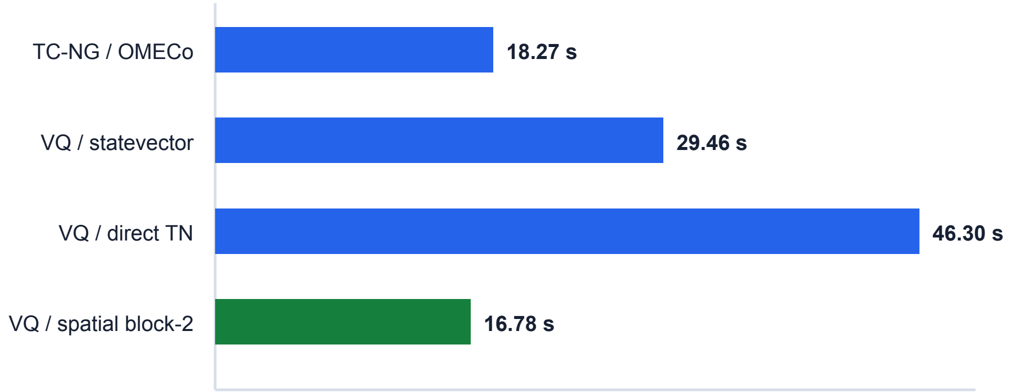
Each candidate runs in an isolated process. Time, host RSS, compiler temporaries, logical residuals, modeled checkpoints, and job-level NVML samples remain distinct fields with explicit semantics.

2. Matched RTX 3090 performance

All rows below use the same physical workload, node, seed, precision policy, and correctness gate. The green row is selected for the declared 100-step objective; the statevector row anchors the current VQETape warm frontier.

Implementation	Compile (s)	First (s)	Warm (ms)	Objective (s)	Host RSS (MiB)	NVML (MiB)
TensorCircuit-NG / OMECo	17.7799	0.2263	2.6577	18.2720	661.3	272
VQETape / statevector	28.7213	0.4025	3.3785	29.4617	462.4	272
VQETape / direct TN	45.6544	0.1376	5.0498	46.2970	575.0	274
VQETape / spatial block-2	15.5816	0.3636	8.3281	16.7781	473.9	274

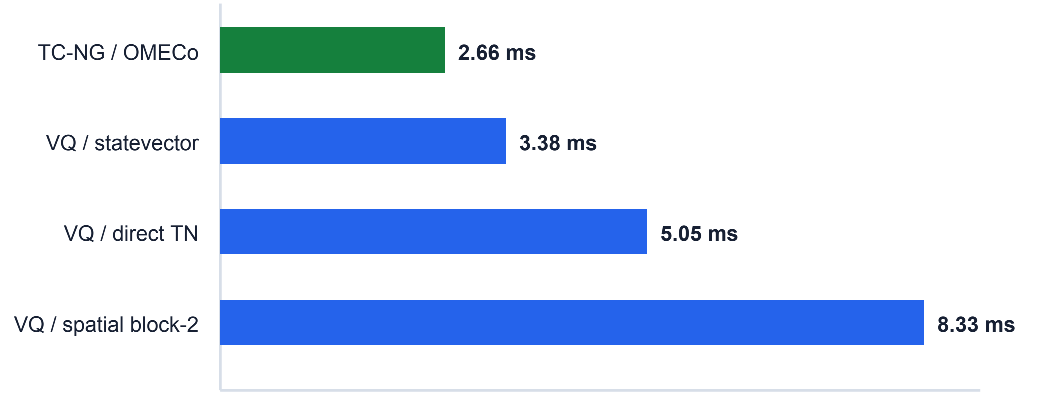
Compile + first + 100 warm (lower is better)



Exact spatial transfer crosses the amortized threshold at 16.7781 s versus 18.2720 s. Compilation is the decisive lever: 15.5816 s versus 17.7799 s.

3. Runtime and memory design space

Warm value-gradient median (lower is better)



Peak host process RSS (lower is better)

NEXT OPTIMIZATION FRONTIER: TensorCircuit-NG records a 2.6577 ms warm reference; the current VQETape statevector frontier is 3.3785 ms (1.27x the reference), while the selected spatial path records 8.3281 ms. This directs the next compiler pass toward warm-kernel fusion while retaining differentiated-program control.

DEMONSTRATED RESULT: host RSS moves from 661.3 to 473.9 MiB. MEASURED TRADE-OFF: job-level NVML samples are 272, 272, 274, and 274 MiB, a compact 272-274 MiB range.

4. Correctness, precision, and provenance

Mathematical and operational comparability is a compiler invariant. VQETape records protocol, seed, precision, value, full gradient, timing boundaries, memory semantics, node/job identity, and source checksums.

Gate	Evidence	Result
Matched correctness	Energy + complete gradient; 1e-5 tolerance	VALIDATED
Full regression	395 tests; 6 declared structural cases	VALIDATED
Incremental suite	17 baseline and Fig. 2 tests	VALIDATED
Static artifacts	27 JSON parse; Python compile; diff check	VALIDATED
TC-NG GPU run	Slurm 23020496; cuda:0; SHA256 audit	VALIDATED
Fig. 2 protocol	RTX 3080 N=6,L=3 execution	VALIDATED
Paper-scale Fig. 2	N=32,L=16 on comparable hardware	SCALE TARGET

Validated Fig. 2 construction

The SU(4) runner preserves 15 x L x (N-1) parameters, TensorNetwork FiniteTFI MPO construction, find/execute separation, slicing controls, and checksum-bound JSON path artifacts. Job 23027373 at N=6, L=3 records energy error 2.38e-07 and gradient relative L2 error 3.29e-07.

SCALE-UP TARGET: execute the same validated protocol at N=32,L=16 on paper-comparable hardware.

5. Four technical innovations

VQETape turns the whole differentiated VQE iteration into a compiler search object.

1 - Differentiated contraction programming

Serialized contraction trees, algebraic transpose einsums, live residual accounting, and checkpoint policies expose the reverse program alongside forward FLOPs and traffic.

2 - Exact spatial-transfer lowering

First/bulk/tail/last programs carry only the exact boundary and avoid a dense transfer matrix. Block width, scan, rematerialization, segmented adjoints, and explicit VJPs become selectable compiler axes.

3 - Commutator-complete adaptive ansatz

Exact insertion gradients and Fubini-Study normalization rank YZ/ZY candidates with contraction-cost deltas. The adaptive circuit reaches 5.05e-11 energy error with 10 parameters; the 14-parameter fixed control records 1.70e-07.

4 - Correctness-gated auto-evaluation

Isolated processes capture timing and distinct memory semantics. A precision bridge maps the declared JAX policy into TensorNetwork's cached backend, making numerical precision a reproducible contract.

JOINT ADVANTAGE: isolated path, AD, checkpoint, optimizer, or ansatz tuning sees only one projection of cost. VQETape can exchange cost across all of them while preserving one exact value-gradient contract.

6. Delivery and research trajectory

Requirement	Status	Reviewer consequence
Repository and harness	DELIVERED	Installable package, CLI/API, tests, safe artifacts
Controlled TC-NG baseline	VALIDATED	Same node, protocol, seed, precision, correctness
Amortized objective	DEMONSTRATED	8.2% improvement at matched size
Host memory	DEMONSTRATED	28.3% lower peak RSS
Warm runtime	NEXT FRONTIER	3.38 ms VQ vs 2.66 ms reference
GPU memory	MEASURED	272-274 MiB job-level range
Paper-scale Fig. 2	SCALE TARGET	Validated protocol ready for N=32,L=16

Clean reproduction

```
python3.12 -m venv .venv
.venv/bin/python -m pip install -e '.[test,baseline]'
.venv/bin/python -m pytest -q
python scripts/build_submission_report.py
```

Reviewer entry points

README.md - thesis, results, and fast-start navigation
submission/vqetape-matched-benchmark.tsv - compact machine-readable rows
submission/submission-status.txt - result map and research frontier
submission/vqetape-technical-report.md - full narrative
submission/report.html - standalone browser report
submission/artifact-manifest.json - SHA256 artifact binding
github.com/JunkaiWang-TheoPhy/issue-33-extreme-efficiency-vqe - public showcase

Research conclusion

VQETape crosses the controlled same-machine baseline threshold on the declared 100-step objective and host RSS while establishing a broader contribution: exact forward, reverse, and physics structure can be compiled together. The warm-kernel reference and N=32,L=16 deployment are now concrete, instrumented optimization targets built on a validated platform.